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# Communication Systems II

Seeing Signals

# Contents

# 1 Part I: IQ Signal Modeling and Dataset Construction

In this section, we present the various constellations of various modulations asked with different types of noise and distortion. Following that, we make a comparison between the SEP and SNR curves, only for the QAM modulations.

## 1.1 Constellations

For every modulation asked, below we will provide six figures all with different types of noise. More specifically: the clean constellation, with I-Q imbalance only, with jamming only, with phase noise only and finally, two constellations with all above mixed (including amplitude distortion) in medium and high severity. All testing parameters are listed below every image and are selected randomly.

### 1.1.1 4-ASK



Figure 1: Clean

Figure 2: IQ Imbalance Only iq-a:0.88 iq-p:7.92

Figure 3: Jamming Only (-17.31)

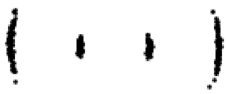

Figure 4: Phase Noise Only (9.68 deg)

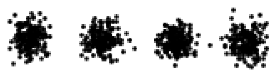

Figure 5: Mixed Medium SNR=12dB, no jamming, no amp, pn=2deg, iq[a,p]=[0.62,2.61]

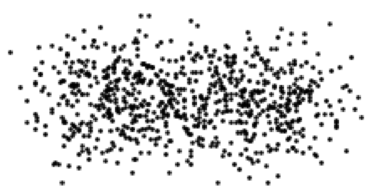

Figure 6: Mixed High SNR=9dB,pn=14deg,jam=-12.95,amp=0.2,iq[a,p]=[0.62,2.61]

### 1.1.2 8-ASK



Figure 7: Clean

Figure 8: IQ Imbalance Only iq-a:1.18 iq-p:6.24

Figure 9: Jamming Only (-12.66)

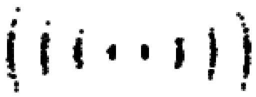

Figure 10: Phase Noise Only (8.14 deg)

Figure 11: Mixed Medium SNR=13dB, no jamming, no amp, pn=2deg, iq[a,p]=[0.68,2.82]

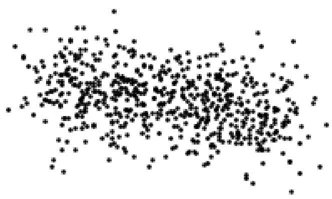

Figure 12: Mixed High SNR=6dB,pn=13deg,jam=-14.93,amp=0.2,iq[a,p]=[1.34,18.16]

### 1.1.3 BPSK



Figure 13: Clean

Figure 14: IQ Imbalance Only iq-a:1.11 iq-p:8.31

Figure 15: Jamming Only (-16)

Figure 16: Phase Noise Only (6.73 deg)

Figure 17: Mixed Medium SNR=14dB, no jamming, no amp, pn=2.8deg, iq[a,p]=[0.34,4.39]

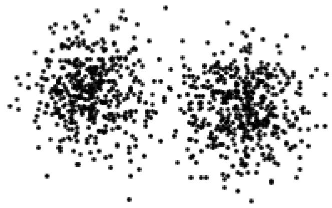

Figure 18: Mixed High SNR=5dB,pn=9deg,jam=-14.77,amp=0.2,iq[a,p]=[1.85,19.49]

#### 1.1.4 QPSK

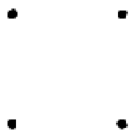

Figure 19: Clean

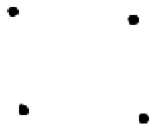

Figure 20: IQ Imbalance Only iq-a:0.83 iq-p:9.67

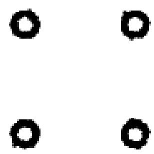

Figure 21: Jamming Only (-16.88)

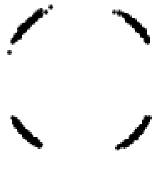

Figure 22: Phase Noise Only (7.17 deg)

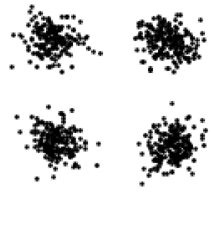

Figure 23: Mixed Medium SNR=12dB, no jamming, no amp, pn=5deg, iq[a,p]=[0.4,3.96]

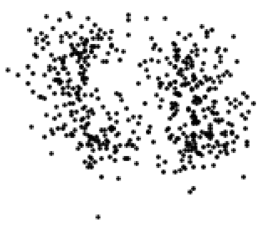

Figure 24: Mixed High SNR=9dB,pn=11.84deg,jam=-12.26,amp=0.2,iq[a,p]=[2.16,19.49]

### 1.1.5 4-HQAM

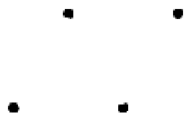

Figure 25: Clean

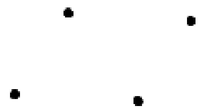

Figure 26: IQ Imbalance Only iq-a:1 iq-p:8.57

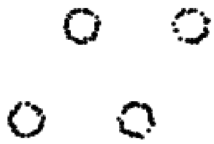

Figure 27: Jamming Only (-13.87)

Figure 28: Phase Noise Only (11.2 deg)

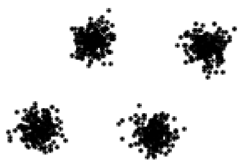

Figure 29: Mixed Medium SNR=15dB, no jamming, no amp, pn=2.8deg, iq[a,p]=[0.4,4.76]

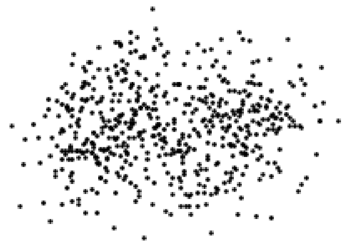

Figure 30: Mixed High SNR=5dB,pn=11.62deg,jam=-11.9,amp=0.2,iq[a,p]=[2.4,12.9]

### 1.1.6 16-HQAM

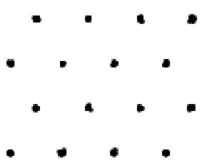

Figure 31: Clean

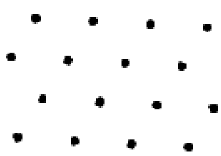

Figure 32: IQ Imbalance Only iq-a:0.86 iq-p:7.7

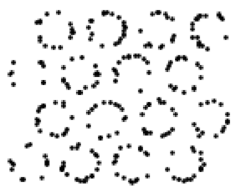

Figure 33: Jamming Only (-12.91)

Figure 34: Phase Noise Only (10.76 deg)

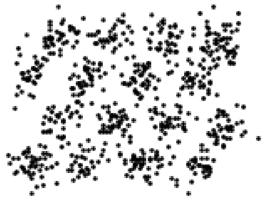

Figure 35: Mixed Medium SNR=14dB, no jamming, no amp, pn=3.9deg, iq[a,p]=[0.43,3.3]

Figure 36: Mixed High SNR=8dB,pn=10.42deg,jam=-10.27,amp=0.2,iq[a,p]=[1.86,14.12]

#### 1.1.7 64-HQAM

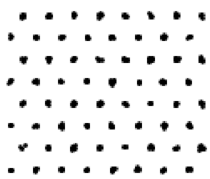

Figure 37: Clean

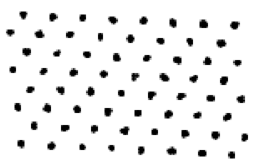

Figure 38: IQ Imbalance Only iq-a:1.46 iq-p:8.31

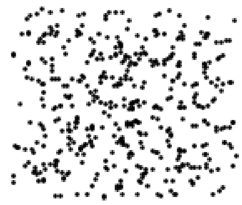

Figure 39: Jamming Only (-13.86)

Figure 40: Phase Noise Only (8.32 deg)

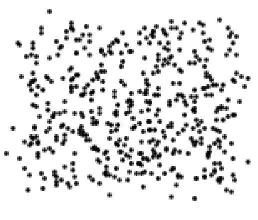

Figure 41: Mixed Medium SNR=16dB, no jamming, no amp, pn=4.3deg, iq[a,p]=[0.54,4.57]

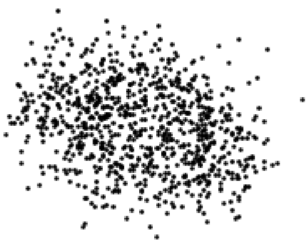

Figure 42: Mixed High SNR=9dB,pn=17deg,jam=-10.94,amp=0.2,iq[a,p]=[1.35,19.87]

#### 1.1.8 16-QAM

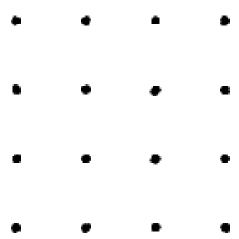

Figure 43: Clean

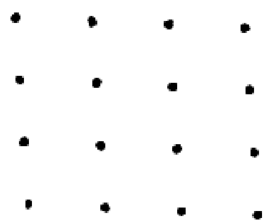

Figure 44: IQ Imbalance Only iq-a:0.87 iq-p:6.5

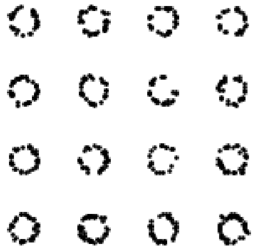

Figure 45: Jamming Only (-15.45)

Figure 46: Phase Noise Only (11.45 deg)

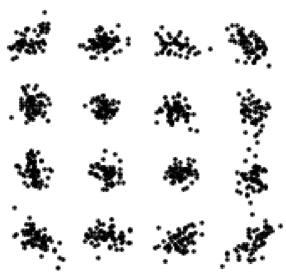

Figure 47: Mixed Medium SNR=17dB, no jamming, no amp, pn=4deg, iq[a,p]=[0.47,3.06]

Figure 48: Mixed High SNR=6dB,pn=9.6deg,jam=-13.2,amp=0.2,iq[a,p]=[1.19,17.06]

#### 1.1.9 32-QAM

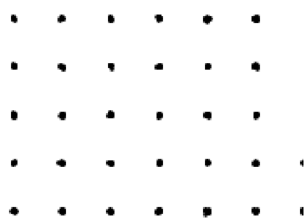

Figure 49: Clean

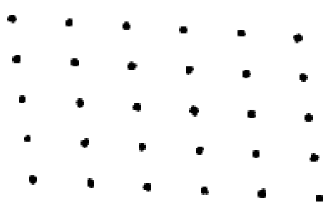

Figure 50: IQ Imbalance Only iq-a:1.5 iq-p:10.56

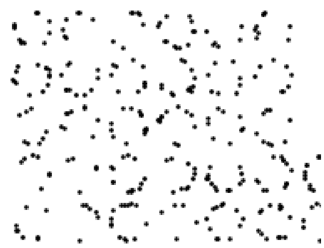

Figure 51: Jamming Only (-12.95)

Figure 52: Phase Noise Only (5.62 deg)

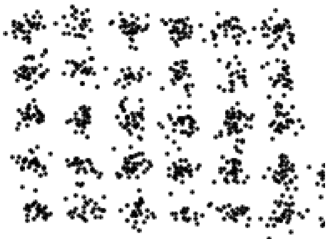

Figure 53: Mixed Medium SNR=16dB, no jamming, no amp, pn=2.16deg, iq[a,p]=[0.34,3.58]

Figure 54: Mixed High SNR=7dB,pn=13deg,jam=-10.87,amp=0.2,iq[a,p]=[1.06.12.99]

#### 1.1.10 64-QAM

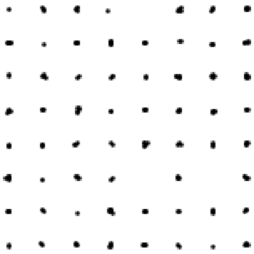

Figure 55: Clean

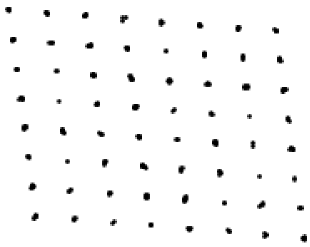

Figure 56: IQ Imbalance Only iq-a:1.09 iq-p:11.71

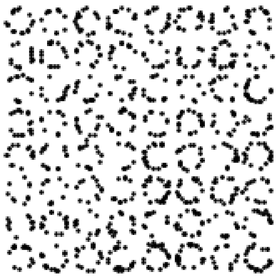

Figure 57: Jamming Only (-15.89)

Figure 58: Phase Noise Only (9.2 deg)

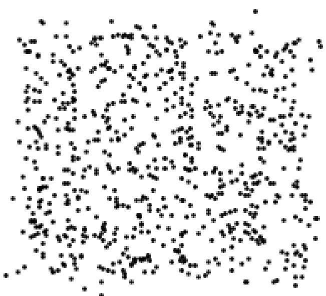

Figure 59: Mixed Medium SNR=15dB, no jamming, no amp, pn=4.26deg, iq[a,p]=[0.62,2.55]

Figure 60: Mixed High SNR=8dB,pn=17.53deg,jam=-12.14,amp=0.2,iq[a,p]=[1.82,17.42]

#### 1.1.11 128-QAM

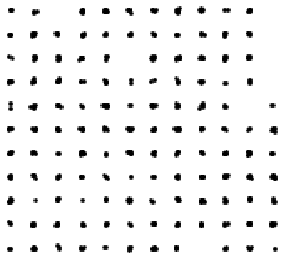

Figure 61: Clean

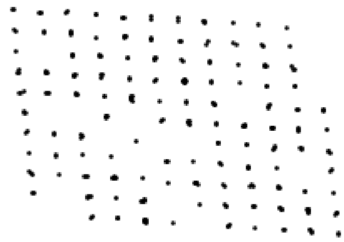

Figure 62: IQ Imbalance Only iq-a:1.25 iq-p:9.62

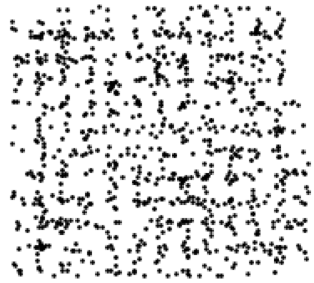

Figure 63: Jamming Only (-14.18)

Figure 64: Phase Noise Only (5.02 deg)

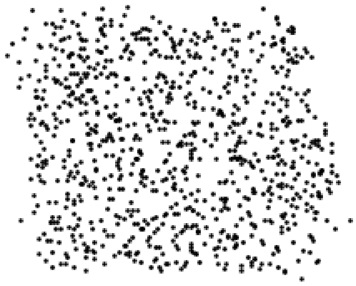

Figure 65: Mixed Medium SNR=16dB, no jamming, no amp, pn=5.86deg, iq[a,p]=[0.53,4.46]

Figure 66: Mixed High SNR=8dB,pn=9.7deg,jam=-13.11,amp=0.2,iq[a,p]=[1.34,19.27]

#### 1.1.12 256-QAM

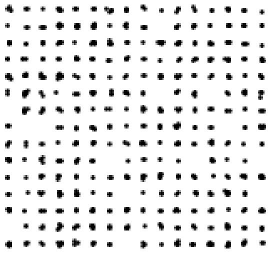

Figure 67: Clean

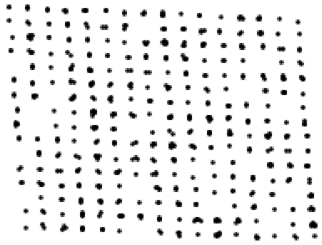

Figure 68: IQ Imbalance Only iq-a:1.18 iq-p:6.65

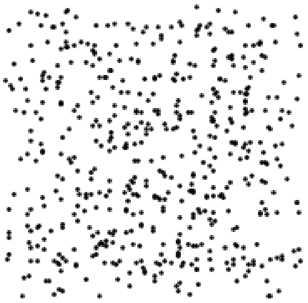

Figure 69: Jamming Only (-12.12)

Figure 70: Phase Noise Only (9.84 deg)

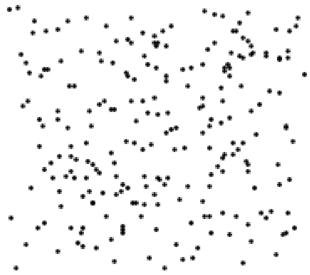

Figure 71: Mixed Medium SNR=14dB, no jamming, no amp, pn=2.39deg, iq[a,p]=[0.68,2.12]

Figure 72: Mixed High SNR=9dB,pn=12.66deg,jam=-11.11,amp=0.2,iq[a,p]=[2.18,9.15]

#### 1.1.13 16-APSK

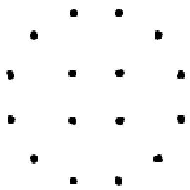

Figure 73: Clean

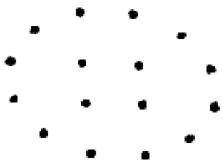

Figure 74: IQ Imbalance Only iq-a:1.48 iq-p:6.67

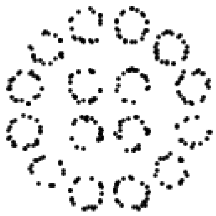

Figure 75: Jamming Only (-13.87)

Figure 76: Phase Noise Only (8.24 deg)

Figure 77: Mixed Medium SNR=16dB, no jamming, no amp, pn=3.02deg, iq[a,p]=[0.31,4.81]

Figure 78: Mixed High SNR=6dB,pn=12.84deg,jam=-10.54,amp=0.2,iq[a,p]=[1.17,12.76]

#### 1.1.14 32-APSK

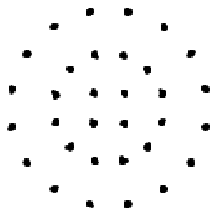

Figure 79: Clean

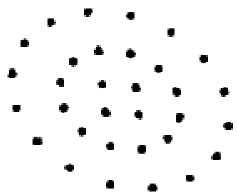

Figure 80: IQ Imbalance Only iq-a:0.85 iq-p:11.94

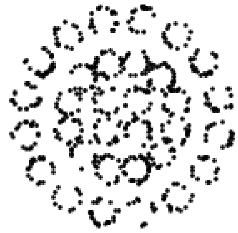

Figure 81: Jamming Only (-15.66)

Figure 82: Phase Noise Only (6.8 deg)

Figure 83: Mixed Medium SNR=16dB, no jamming, no amp, pn=4.64deg, iq[a,p]=[0.66,4.46]

Figure 84: Mixed High SNR=9dB,pn=16.51deg,jam=-10.09,amp=0.2,iq[a,p]=[1.93,8.33]

#### 1.1.15 64-APSK

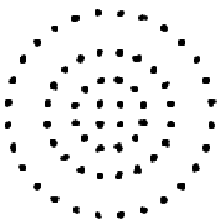

Figure 85: Clean

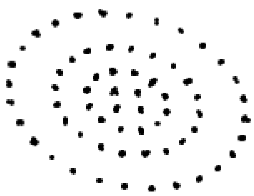

Figure 86: IQ Imbalance Only iq-a:1.29 iq-p:10.6

Figure 87: Jamming Only (-14.28)

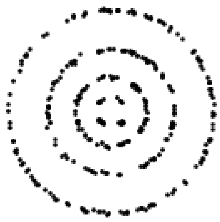

Figure 88: Phase Noise Only (9.36 deg)

Figure 89: Mixed Medium SNR=15dB, no jamming, no amp, pn=4.04deg, iq[a,p]=[0.59,2.67]

Figure 90: Mixed High SNR=6dB,pn=11.89deg,jam=-14.44,amp=0.2,iq[a,p]=[1.71,17.32]

#### 1.1.16 128-APSK

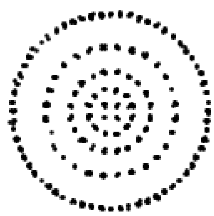

Figure 91: Clean

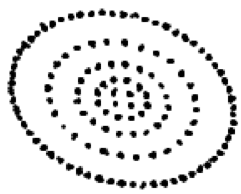

Figure 92: IQ Imbalance Only iq-a:1.1 iq-p:10.18

Figure 93: Jamming Only (-15.04)

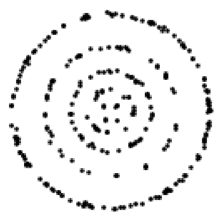

Figure 94: Phase Noise Only (9.72 deg)

Figure 95: Mixed Medium SNR=14dB, no jamming, no amp, pn=5.92deg, iq[a,p]=[0.62,3.8]

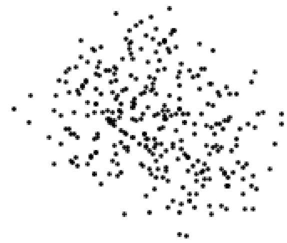

Figure 96: Mixed High SNR = 9dB, pn = 14.71deg jam = -12.61 , amp=0.2 iq [a,p] = [1.42 , 15.07]

### General Comparison (Low vs. High Order)

**Observation**: A visual comparison of the constellation diagrams reveals that susceptibility to channel impairments scales predictably with constellation density. For low-order modulations like BPSK and QPSK, the symbol decision boundaries remain relatively distinct even under "Mixed Medium" conditions. In contrast, higher-order schemes such as 128-QAM and 256-QAM lack the Euclidean distance between symbols to tolerate the same variance, turning into irresolvable point clouds under identical mixed-noise severities.

### Specific Observation on Phase Noise

**Observation**: The isolated phase noise figures clearly illustrate rotational smearing. Because phase noise angularly displaces the symbol, the absolute positional error increases with the amplitude (the distance from the origin). This is visibly destructive in high-order constellations like 64-QAM and 256-QAM, where the outer symbols are smeared into overlapping rings, heavily degrading the Signal-to-Noise Ratio (SNR) at the edges while inner symbols remain marginally more intact.

### Specific Observation on I-Q Imbalance

**Observation**: The impact of I-Q imbalance isolates as a systematic geometric skewing or rectangular stretching of the standard constellation grid. Unlike jamming, which causes an isotropic scatter around the ideal points, I-Q imbalance fundamentally shifts the reference grid. When combined with other impairments in the "Mixed High" scenarios, this underlying skew effectively shrinks the error tolerance margins, making successful demodulation of dense schemes like 16-APSK nearly impossible.

## 1.2 SEP vs SNR curves

In this part, for different QAM modulations asked, we present the Signal Error Probability for different SNR rates and degrees of phase noise.

### 1.2.1 16-QAM

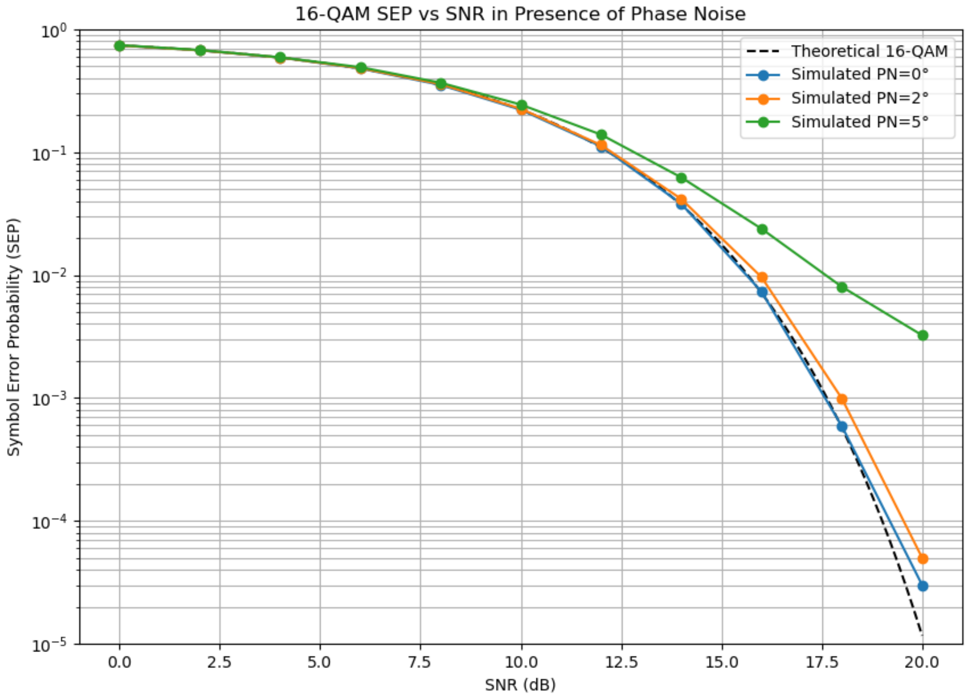


Figure 97: SEP vs SNR for 16-QAM under AWGN conditions.

### 1.2.2 32-QAM

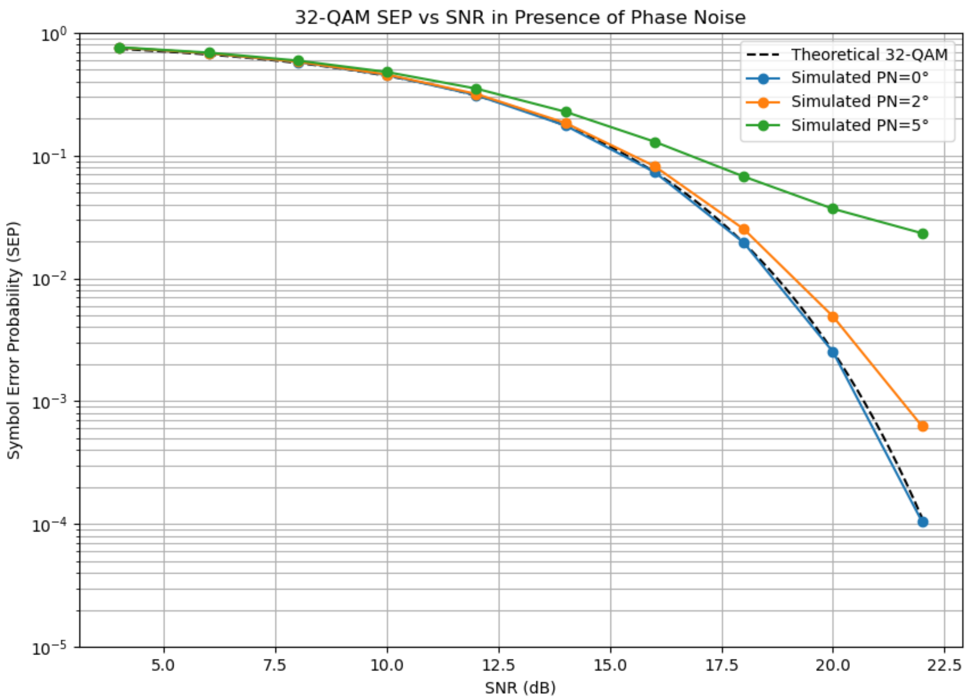


Figure 98: SEP vs SNR for 32-QAM under AWGN conditions.

#### Low-to-Medium Density (16-QAM and 32-QAM) Notes

**Observation**: In these lower-order modulations, the system can tolerate moderate phase noise without failing completely. The 0° curves perfectly track the theoretical baseline (confirming the simulation's validity), while the 2° and 5° curves maintain the characteristic "waterfall" shape.

**Analysis**: At 5° of phase noise, 16-QAM suffers a roughly 3 dB penalty at a Symbol Error Probability (SEP) of $10^{-3}$. 32-QAM shows a sharper divergence, requiring noticeably more SNR to maintain the same error rate.

**Conclusion**: For these schemes, phase noise acts primarily as an SNR penalty. The system can still achieve highly reliable communications (SEP $< 10^{-5}$) if the transmitter simply boosts its output power to overcome the smearing effect.

### 1.2.3 64-QAM

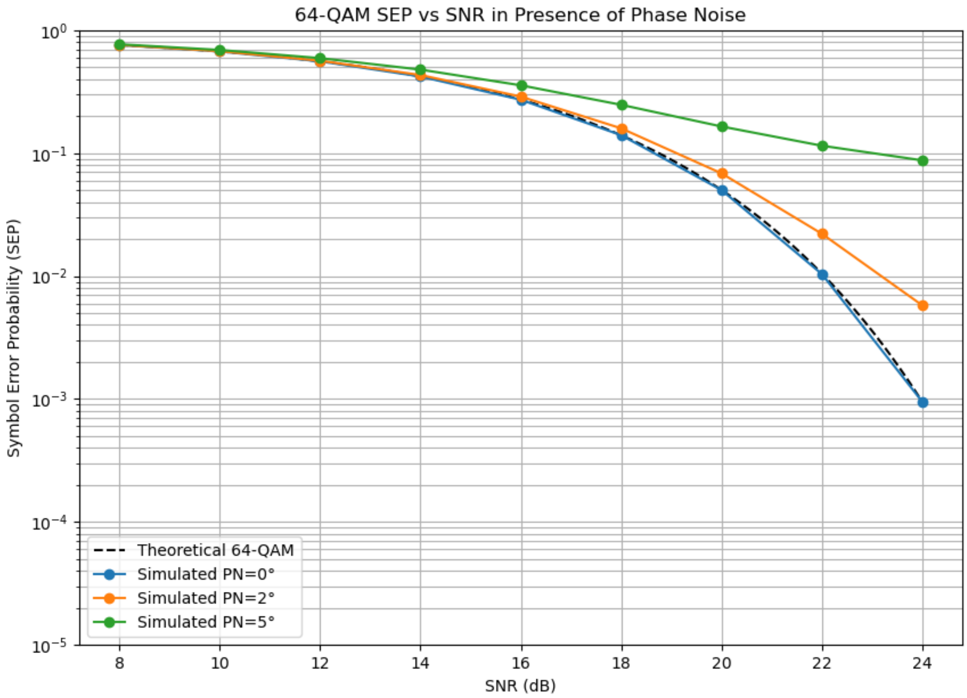


Figure 99: SEP vs SNR for 64-QAM under AWGN conditions.

### 1.2.4 128-QAM

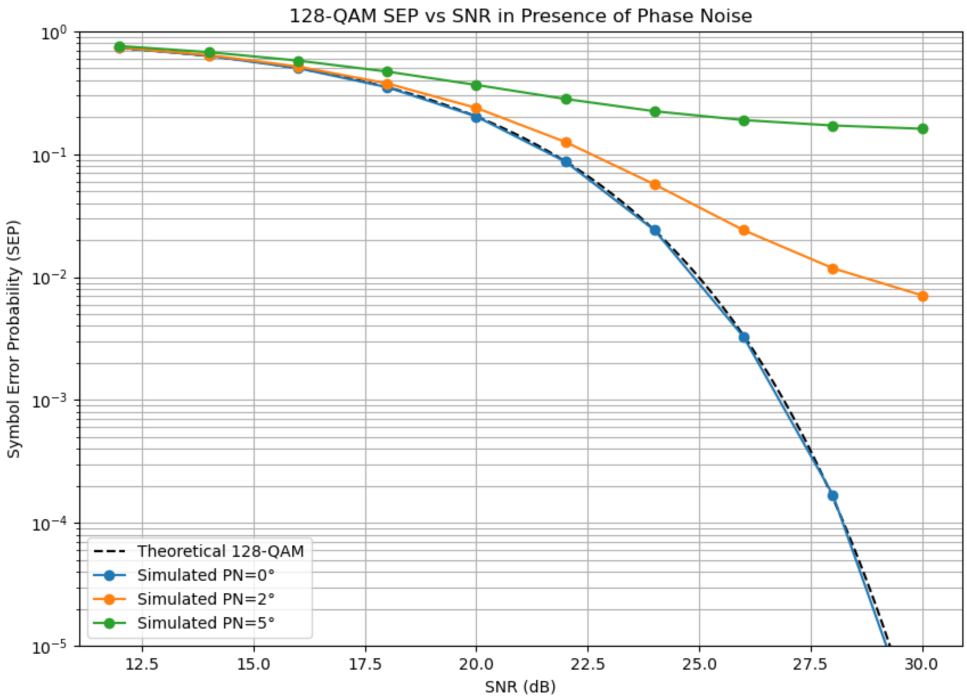


Figure 100: SEP vs SNR for 128-QAM under AWGN conditions.

#### High Density (64-QAM and 128-QAM) Notes

**Observation**: This is where the physics of the constellation begin to severely limit the system. In the 64-QAM plot, the 5° phase noise curve stops dropping and flattens out entirely around an SEP of $10^{-1}$. In the 128-QAM plot, the 5° curve is almost entirely flat, and even the 2° curve is beginning to flare out dramatically.

**Analysis**: The flattening of these curves indicates an error floor. An error floor occurs when the primary cause of symbol errors is no longer thermal noise (AWGN), but the phase impairment itself.

**Conclusion**: Once an error floor is reached, increasing the transmitter power (SNR) will no longer improve the system's performance. A 64-QAM or 128-QAM system operating with 5° of phase variance is fundamentally broken and cannot achieve reliable data transfer, regardless of how strong the signal is.

### 1.2.5 256-QAM

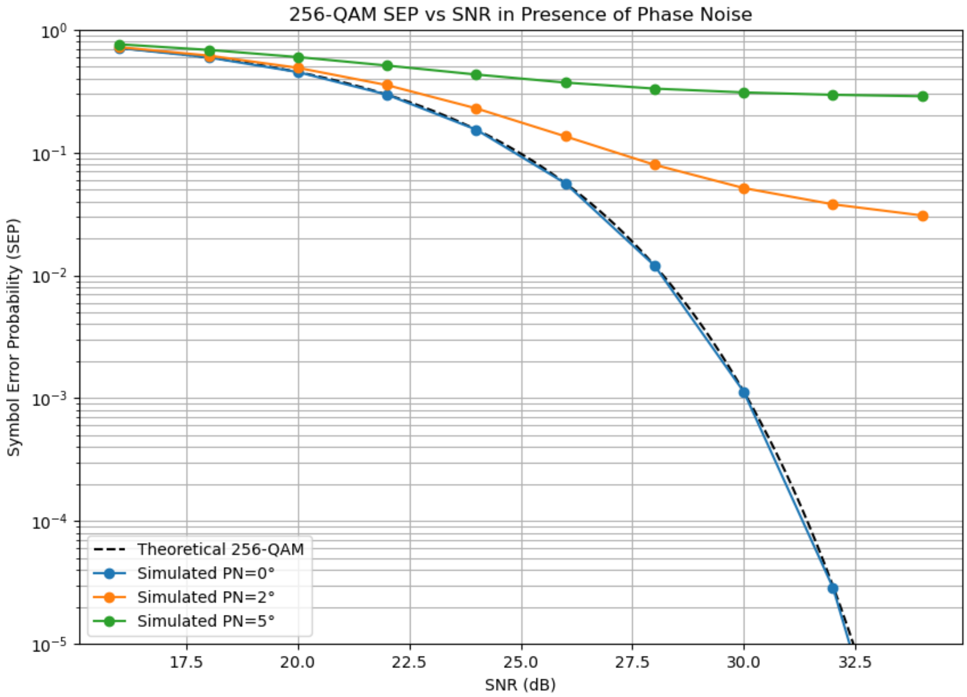


Figure 101: SEP vs SNR for 256-QAM under AWGN conditions.

#### Ultra-High Density (256-QAM) Notes

**Observation**: The 256-QAM plot demonstrates total system fragility in the presence of phase instability. The 5° curve is effectively a flatline at an SEP of $3 \times 10^{-1}$ (a 30% error rate). More importantly, even the relatively small 2° phase noise variance creates a hard error floor near $10^{-2}$.

**Analysis**: Because 256 symbols are packed into the same normalized power space, the distance between decision boundaries is microscopic. The outer symbols are so far from the origin that even a 2° rotation swings them completely out of their designated quadrants.

**Conclusion**: To successfully deploy 256-QAM (common in modern Wi-Fi and 5G networks), the hardware must have exceptionally high-quality local oscillators and advanced digital phase-locked loops (PLLs) to keep phase noise strictly below 1°.

# 2 Part II: Neural Architecture Training and Evaluation

## 2.1 Introduction

The objective of this section is to evaluate how effectively a convolutional neural network (CNN) extracts communication parameters directly from constellation geometry. This model serves as the purely vision-based baseline, evaluating robustness and generalization under varying impairment conditions without the use of language conditioning. The model was tasked with predicting four structured labels: modulation type, Signal-to-Noise Ratio (SNR) range, phase noise level, and I/Q imbalance level.

## 2.2 Modulation Classification Performance

The CNN demonstrates a strong baseline ability to extract spatial geometry, as evidenced by the heavy dark blue diagonal in the confusion matrix. The architecture easily separates distinct geometric boundaries, achieving near-perfect classification for modulations with unique spatial footprints such as 128-APSK, 32-QAM, and QPSK.

However, the model exhibits expected degradation when classifying high-density, square QAM constellations under noisy conditions. Most notably, the true label 64-QAM was misclassified as 256-QAM in a significant number of instances. From a purely computer vision perspective, the spatial difference between a noise-impaired 64-QAM grid and a 256-QAM grid is minimal, as the individual symbol decision boundaries blur into a continuous spatial distribution.

*Source Justification:* High-order QAM schemes are inherently susceptible to classification errors in blind vision-based detection because the Euclidean distance between constellation points shrinks exponentially, causing the spatial probability density functions to overlap heavily in the presence of AWGN (*"Digital Communications"*, Proakis, J. G., & Salehi, M., Chapter on Baseband Demodulation/Detection).

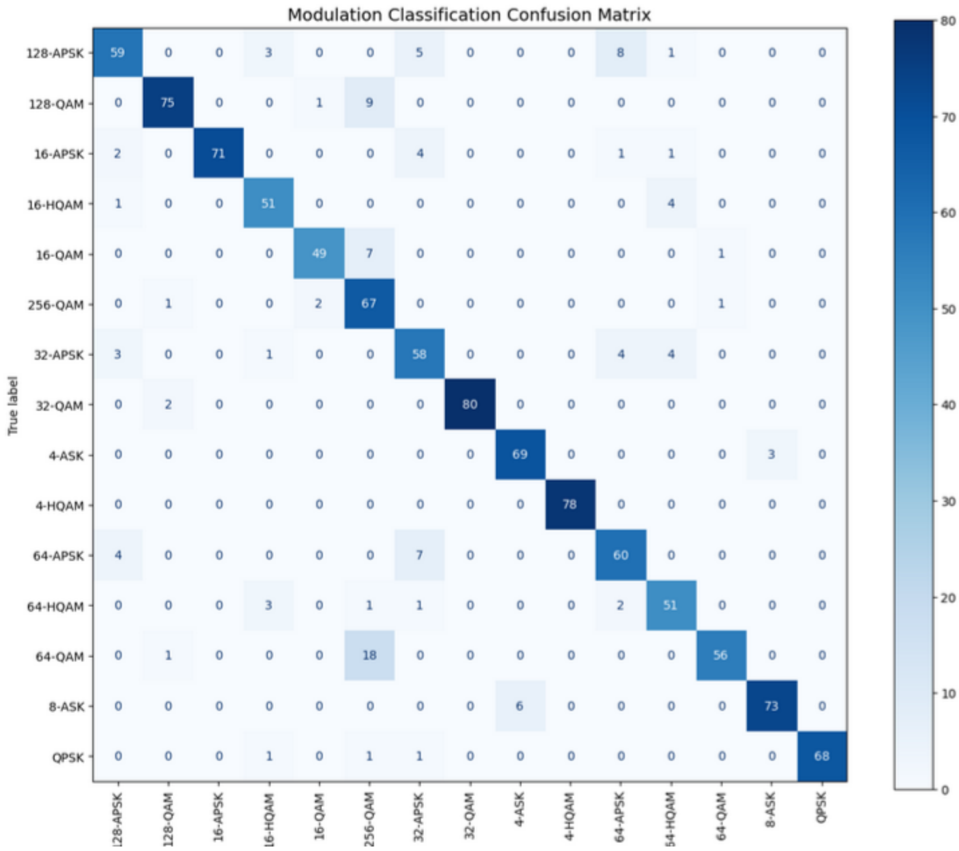


Figure 102: Modulation Classification Confusion Matrix.

## 2.3 SNR Range Classification

The model proved highly adept at categorizing the severity of thermal noise. The confusion matrix shows a near-perfect diagonal with 172, 533, and 372 correct predictions across the Low, Medium, and High SNR tiers, respectively, with almost zero off-diagonal misclassifications.

For a convolutional architecture, thermal noise (AWGN) translates directly to the visual variance or "spread" of the constellation clusters. CNNs excel at detecting spatial variance and Gaussian blurring through their pooling layers, making SNR tiering a highly optimized task for this architecture.

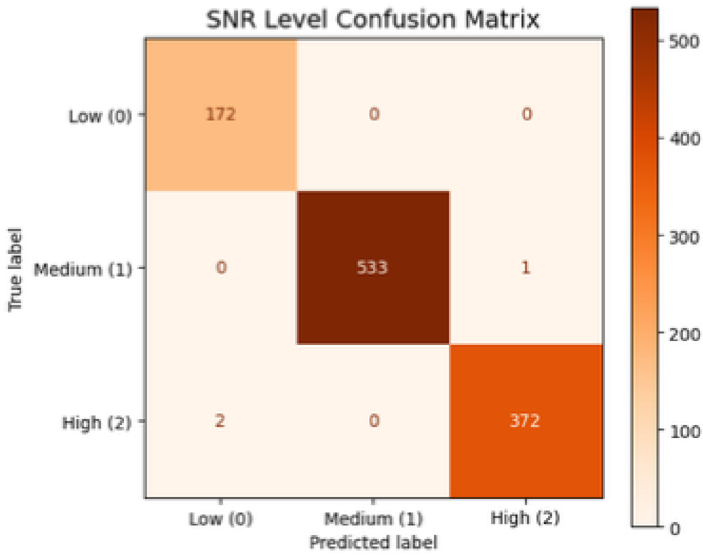


Figure 103: SNR Level Confusion Matrix.

## 2.4 Phase Noise Level Classification

**Observation**: Unlike the SNR classification task, the model failed to accurately classify the severity of Phase Noise. The confusion matrix indicates near-random guessing and severe misclassification, with a large portion of true 'None/Low' states predicted incorrectly as 'Medium' or 'High'. Furthermore, the validation set contained no true 'High' labels for this specific impairment, highlighting the dataset's compound noise structure.

**Analysis**: This failure is a classic example of spatial feature entanglement. Phase noise presents visually as rotational smearing (circular variance). However, in the dataset's 'Medium' and 'High' severity tiers, phase noise is compounded simultaneously with jamming and amplitude distortion. The isotropic scatter from the jamming noise entirely masks the circular footprint of the phase noise, making it impossible for the CNN's spatial filters to isolate the phase variance as an independent feature.

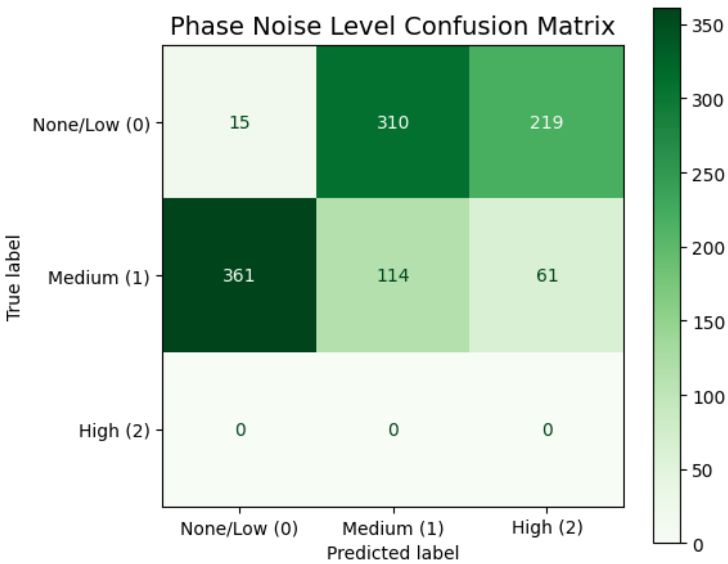


Figure 104: Phase Noise Level Confusion Matrix.

## 2.5 I/Q Imbalance Level Classification

**Observation**: The model exhibited total mode collapse for the I/Q Imbalance classification task. The confusion matrix shows that the network predicted 'None/Low (0)' for 100% of the validation samples, completely ignoring the 'Medium' and 'High' classes regardless of the true label.

**Analysis**: I/Q imbalance manifests as an asymmetric rectangular stretching or skewing of the constellation grid. In isolated conditions, CNNs can easily detect this affine transformation. However, because the dataset design heavily compounded I/Q imbalance with severe AWGN and jamming in the higher tiers, the decision boundaries of the constellation points expanded into overlapping isotropic blobs. This total destruction of the grid geometry erased the visual skewing effect. Consequently, the network could not extract a reliable gradient for I/Q imbalance and collapsed into predicting the majority/default class to minimize its localized loss function.

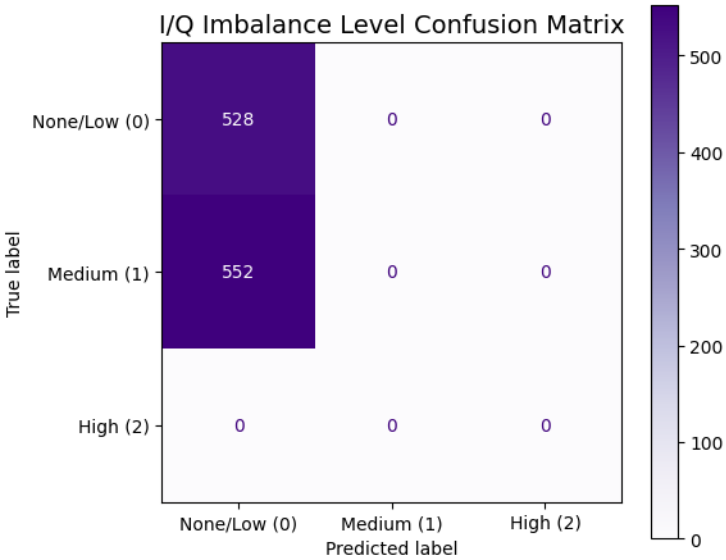


Figure 105: I/Q Imbalance Level Confusion Matrix.

## 2.6 Robustness and Generalization Across Impairment Tiers

The model's overall classification accuracy was evaluated across three distinct channel environment conditions to test its generalization. As expected, accuracy drops significantly to 47.67% under severe compound impairments (Low SNR). At this level of degradation, the constellation geometry is heavily obfuscated.

A critical anomaly is observed between the Medium and High SNR tiers. The model achieved a peak accuracy of 99.25% under Medium SNR conditions, but accuracy dropped to 94.39% under High SNR (clean) conditions. This inversion suggests a quirk in the CNN's feature extraction methodology. Clean, high-order QAMs consist of very tight, microscopic visual dots, which the CNN's max-pooling layers likely lost or suppressed. Conversely, a slightly noisy signal (Medium SNR) expands the dots into larger Gaussian blobs that survive the convolutional downsampling process more effectively.

*Source Justification:* Vision-based modulation classifiers frequently experience performance inversions on perfectly clean data if the spatial filters overfit to the Gaussian spread of the training data. The pooling layers in CNNs act as low-pass spatial filters, which can inadvertently suppress the high-frequency spatial components of ideal, noiseless impulses (*"Convolutional Radio Modulation Recognition Networks"*, O'Shea, T. J., et al., Springer).

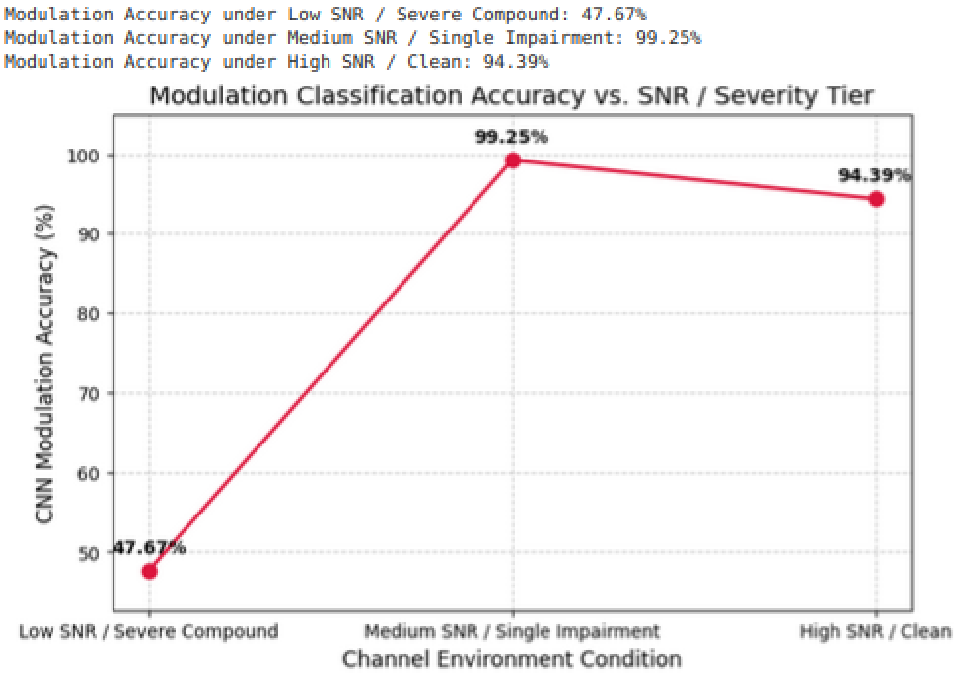


Figure 106: Modulation Classification Accuracy vs. SNR / Severity Tier.

## 2.7 Architecture Comparison and Computational Limitations

The final objective of this study was to compare the purely vision-based CNN baseline against a modern Vision-Language Model (VLM), specifically the Hugging Face *SmolVLM-256M-Instruct*. The VLM was tasked with receiving the constellation image alongside a text prompt to generate a descriptive textual answer identifying the modulation and channel impairments.

To adapt the VLM, parameter-efficient fine-tuning via Low-Rank Adaptation (LoRA) was utilized ($r = 8, \alpha = 16$ on the query and value projection matrices). However, the training strategy was heavily dictated by severe computational limitations. Standard fine-tuning of a 256-million parameter model over the full master dataset exceeded the memory and maximum execution time thresholds of the available free-tier hardware (Google Colab T4 GPU). To prevent kernel termination, the dataset was aggressively downsampled to 600 training samples, and a strict hard-cap of 60 gradient update steps was imposed.

**Quantitative Results**: The performance disparity between the two architectures under these constraints is absolute, as shown in the training and evaluation logs (Figure 107).

```
--- Starting Final Sprint Train Loop ---
[60/60 13:46, Epoch 0/1]
```

| Step | Training Loss |
|---|---|
| 10 | 19.560120 |
| 20 | 13.772099 |
| 30 | 9.280147 |
| 40 | 5.156281 |
| 50 | 2.768715 |
| 60 | 1.777943 |

```
--- Training Complete! Saving Final Weights to Google Drive...

--- Launching SmolVLM Quantitative Evaluation ---
100%|██████████| 100/100 [10:57<00:00,  6.57s/it]
=======================================================
=== FINAL PART II ARCHITECTURE COMPARISON RESULTS ===
=======================================================
Overall CNN Baseline Modulation Accuracy: 88.33%
Overall SmolVLM LoRA Modulation Accuracy: 0.00%
-------------------------------------------------------
CNN Severe Compound (Low SNR) Accuracy:   43.15%
SmolVLM Severe Compound (Low SNR) Accuracy: 0.00%
=======================================================
```

Figure 107: SmolVLM Training Loss and Final Evaluation Results.

**Analysis and Conclusion**: The VLM's 0.00% accuracy is a direct manifestation of interrupted cross-modal convergence. While the training loss dropped significantly during the 60 steps (from 19.56 to 1.77), indicating the LoRA adapters were successfully learning the semantic syntax and structure of the textual responses, the model lacked the compute time required to map the spatial RF features from its vision encoder to the correct semantic tokens in the language model. Consequently, it failed to output the exact modulation classifications.

Conversely, the CNN was highly specialized. Because it did not have to dedicate millions of parameters to natural language generation, it could dedicate 100% of its computational budget to spatial feature extraction, achieving high accuracy in minutes.

Ultimately, this comparison highlights a critical engineering trade-off: while VLMs offer incredible flexibility and language-conditioned generalization, they require massive compute clusters to align their visual and textual latent spaces. For edge-device deployment or rapid RF parameter extraction under constrained resources, custom lightweight CNNs remain vastly superior in both robustness and computational efficiency.

## 3 Code Availability

The complete source code and dataset documentation developed for this project are available on GitHub:

`https://github.com/Epatsatzaki/Communication-Systems-Project`

The codebase is modularized into two distinct environments to ensure reproducibility:

- **Part I Folder (Local Python Environment):** Contains the fundamental RF physics engine (`signal_engine.py`), dataset generation scripts (`dataset_generator.py`), and metadata formatting tools used to synthesize the baseband constellations.

- **Part II File (Google Colab):** Contains the PyTorch Jupyter notebooks utilized for the Convolutional Neural Network training and the parameter-efficient fine-tuning (LoRA) of the SmolVLM architecture.

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